THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL UNIT TEST SET 3

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The sum $1^3 + 2^3 + \cdots + 9^3$ equals: 1

(a) 2024 **(b)** 2070 **(c)** 2025 **(d)** 2026

2. Rays OA, OB, OC, OD make equal angles around O (a full turn). The measure of each angle is: 1

(a) 82 **(b)** 62 **(c)** 72 **(d)** 162

3. On reversing and adding 187, the first sum is: 1

(a) 979 **(b)** 978 **(c)** 968 **(d)** 957

4. A pictograph has symbols worth 12 each. A value of 100 needs how many full symbols and a partial one? Number of full symbols: 1

(a) 10 **(b)** 8 **(c)** 9 **(d)** 7

5. The sum of the first 3 prime numbers after 10 is: 1

(a) 42 **(b)** 40 **(c)** 41 **(d)** 43

6. The 41th triangular number minus the 40th triangular number is: 1

(a) 40 **(b)** 41 **(c)** 42 **(d)** 43

7. A wheel of 12 equal spokes: the angle between two neighbouring spokes is 30°. The angle between spoke 1 and spoke 6 (going one way) is: 1

(a) 140 **(b)** 150 **(c)** 160 **(d)** 180

8. 89 is added to its reverse and the sum is 187. How many reverse-and-add steps in all are needed to reach a palindrome? 1

(a) 21 **(b)** 22 **(c)** 20 **(d)** 19

9. In a bar graph with scale 1 cm = 8 units, the value 60 is shown by a bar of height (cm): 1

(a) 8 **(b)** 8.5 **(c)** 7.5 **(d)** 6.5

10. The largest prime number less than 100 is: 1

(a) 96 **(b)** 97 **(c)** 99 **(d)** 98

11. In a party 18 people greet each other once with a hug. Number of hugs: 1

(a) 171 **(b)** 153 **(c)** 135 **(d)** 154

12. At 2:30 the smaller angle between the clock hands is: 1

(a) 100 **(b)** 120 **(c)** 105 **(d)** 110

13. How many steps does the Kaprekar routine take to reach 6174 from 5273? 1

(a) 6 **(b)** 7 **(c)** 8 **(d)** 9

14. The total of the table, after removing the largest value, is: 1

Item	Number
Guava	70
Grapes	70
Orange	40
Mango	70
Apple	60
Banana	25

(a) 260 **(b)** 275 **(c)** 270 **(d)** 265

15. The product of the two smallest prime factors of 330 is: 1

(a) 6 **(b)** 8 **(c)** 7 **(d)** 5

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the sum of odd numbers from 19 to 47. 2

17. 8 rays start from O so that every pair makes an angle. Find the number of angles formed. 2

18. A number has 12 digits, all equal to 5. Find its digit sum and the remainder when the number is divided by 9. 2

19. The data shows number of goals in 6 matches: 2, 3, 1, 4, 0, 5. Find the range and the mean. 2

20. Find the prime factors of 84 and the smallest number to multiply to make it divisible by 45. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the smallest number which when divided by 6, 8 and 12 leaves remainder 5 each time. 3

22. In a pattern the 1st term is 23, the 2nd is 30, and so on by adding 7. Which term of the pattern equals 331? Find also the term just before it. 3

23. At 3:10 how far (in degrees) is the minute hand ahead of the hour hand, and how far is it from 12? 3

24. Find a 4-digit palindrome that is divisible by 11 and has digit sum 20. Verify. 3

OR

The table shows two weeks of sales (units). Find the week with the larger total and by how much. 3

Day	Week 1	Week 2
Mon	14	15
Tue	20	20
Wed	13	14
Thu	17	13

25. Three traffic lights change every 30, 45 and 60 seconds. If they change together at 8:00 am, after how many seconds do they next change together and at what time? 3

OR

Sums of consecutive odd numbers give squares. Find the sum of the odd numbers from 1 to 49 and use it to find 25^2. Then evaluate the sum from 51 to 61. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. 100 students were asked their favourite colour: red 20, blue 25, green 15, yellow 10 and the rest chose pink. Find the number liking pink, the fraction liking blue, and draw the pictograph with one symbol = 5 (give symbol counts). **5**

OR

Find the smallest 4-digit number divisible by 6, 8 and 12. Also find the largest 4-digit number divisible by all three. **5**

27. Look at the pattern: the 6th figure has 50 dots. How many dots are in figure 15? If a figure has 176 dots, which figure is it? Show that the dots form a pattern with common difference 9. **5**

OR

At 2:50, find **5**

(a) the angle turned by the hour hand since 12 o'clock,

(b) the angle turned by the minute hand since 12,

(c) the smaller angle between the hands,

(d) its type.

END OF QUESTION PAPER — Total: 50 Marks

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